Disclosure Day

Compact proof that we most likely inhabit an OPH simulation

Paper release: r1519 Released: July 8, 2026

OPH makes a falsifiable compression claim: five axioms plus two closure constants generate the observer-facing structure of our Universe, including many measured quantities, without using those measurements as inputs. A single lucky hit would mean little; OPH asks the same fixed branch to produce quantum readout, Standard Model integer structure, a source-side no-hadron fine-structure near-endpoint, a charged-family shape, the QCD-free weak hierarchy, and geometric gravity normalization.

Once the rules and readout maps are fixed before comparison, the finite branch has one output. OPH is wrong if that output leaks targets, admits an equally valid branch, misses the measured structures it claims to generate, or fails the stated tests. Numbers like $\alpha(0)$, SI G, hadronic inputs, PDG tables, cosmology scorecards, hardware lanes, and bridge-defined capacities are checks and failure points until OPH derives them without using the measured answer.

What this proof contributes

The proof gives one readable path through OPH. Companion papers carry the longer derivations. Standard tools keep their usual meanings. OPH adds finite observer-patch closure, $P_{\mathrm{src}} = P_{\mathrm{cand}}$, $N_\star$, and the rule that a row counts only when the declared map emits it. That small source has to carry observer patches, quantum readout, Standard Model branch integers, the source-side fine-structure prediction, charged-family shape, weak hierarchy, gravity normalization, and failure tests in one argument.

Anti-fitting rule

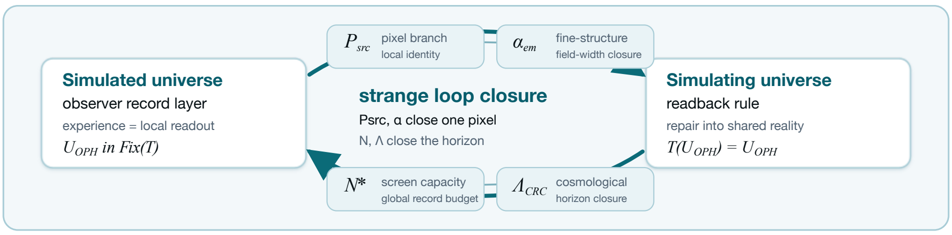
A numerical row is counted in our argument only if the paper or code gives four things: the calculation that emits it, no hidden path from the measured answer into that calculation, an interval or uniqueness check, and a clear way for the row to fail. A measurement may identify the basin to inspect. The endpoint must come from the source map. If the target value, or an algebraically equivalent calibrated proxy, appears in the calculation path, the row is comparison data, a display check, empirical closure, or a test surface. It sits outside the compact anti-circularity proof.

Most people picture a simulation as a machine that renders a world frame by frame. OPH uses a different mechanism: a fixed-point computation without a global timeline.

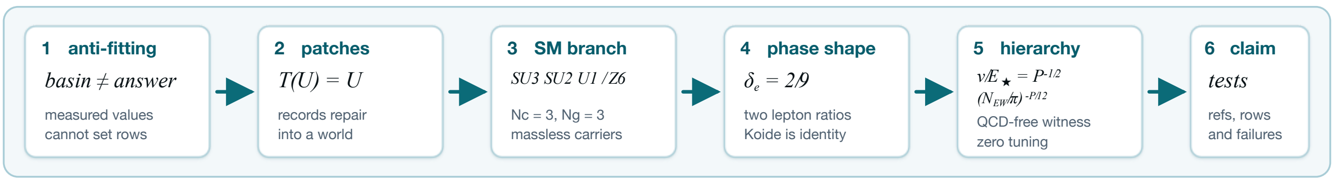
$$\text{ordinary simulation: } U_t \to U_{t+1} \to U_{t+2} \to \cdots, \qquad \text{OPH: } \mathcal{T}(W) = W.$$

The symbol W is the globally self-consistent observer-readable structure. $\mathcal{T}$ is the OPH readback-and-repair operator over observer patches, records, overlaps, mismatch scores, repair moves, and checkpoint continuation [O1,O2,O3]. The universe is the normal form of that computation. From inside, observers read it as history and experience that reading as time. Hofstadter's name for this kind of thing is a strange loop: a "paradoxical level-crossing feedback loop" [S16]. In OPH, strange loops sit at the core of the finite observer construction. Measurements may point to the right branch. Their role is branch localization. The counted rows below use that rule for the charged-family shape and hierarchy witnesses. The remaining rows are comparison data or falsification surfaces with declared claim classes.

Bracketed source keys matter. When this note names a theorem without printing the full proof, the numbered OPH paper carries the proof details. The direct reading anchors at the end point to the online source lines. For fast lookup: OPH axioms are [A1], consensus repair is [A2], pixel closure is [A4], capacity closure is [A5], the QCD-free hierarchy witness is [A10], and the Standard Model gauge quotient is [A21]. The derivation chains printed here and in the referenced papers should be sound and complete; if you find an error, contact info@floatingpragma.io.



The labels name what the values do: P_{src} fixes the local pixel branch used in the compact witnesses, α is the field-width readback, $N_\star$ fixes total screen capacity, and Λ is the horizon closure seen in cosmology.



OPH's five axioms are listed in *Observers Are All You Need* [O1]. Think of them as the toolchain. The argument below uses the finite observer-patch normal form, the exact quantum event surface, the source pixel coordinate, the global capacity target, the recovered Standard Model branch, and one compact three-scalar cross-lock:

$$(N_c, N_g, \beta_{\mathrm{EW}}) = (3, 3, 4) \quad \Longrightarrow \quad \left(\frac{v}{E_\star}, \frac{m_\mu}{m_e}, \frac{m_\tau}{m_e}\right).$$

The table says which rows are counted in this compact argument and which rows stay comparison data or falsification targets [O1,O3,O4,O5,O7,O8,O9].

1. The fixed-point computation

In OPH, "simulation" has a stricter meaning than "fixed point." The theorem-level test is the OPH simulation test [O2,A23]. It requires five independent properties: endogenous update, observer-readable records, overlap repair with a schedule-independent normal form, selected-branch elimination, and implementation/clock closure. The mathematical object is a finite patch system. Patches expose ports, write records, compare overlaps, repair disagreement, and return a shared normal form after readback [O1,O2,O3]. Internal observers experience that normal form as the universe. The system consists of finite patches, shared overlap ports, recovery-derived repair maps, checkpoints, and a stable world read from inside [O1,O2,O3].

The symbol $\mathfrak{U}_{\mathrm{OPH}}$ denotes the selected OPH universe. The map T is the observer-overlap repair map [O1,O2]. The notation $\mathrm{Fix}(T)$ means the states returned unchanged by that map:

$$T(\mathfrak{U}_{\mathrm{OPH}}) = \mathfrak{U}_{\mathrm{OPH}}.$$

The fixed-point equation follows from the five-clause simulation test. The proof in [O2,A23] first checks those clauses on the declared finite branch and then derives the identity. Ordinary equilibria fail this test; the same reference gives simple counterexamples and lists the failure modes. This note cites that proof instead of repeating it.

Conditional OPH closure theorem

Fix OPH axioms, branch rules, and quotient/readout maps before test data, and require source maps to be forced by those rules up to observer-invisible equivalence. Let

$$\Gamma_P : I_P \to I_P, \qquad \Gamma_N : I_N \to I_N$$

be source-only contractions. Suppose their dependency graphs contain no held-out target path, no admissible source-map deformation survives the axioms and branch-selection rules, and the finite packet passes the simulation test of [O2,A23]. Then

$$P_\star = \Gamma_P(P_\star), \qquad N_\star = \Gamma_N(N_\star)$$

have unique solutions. Every declared quotient observable

$$Y_i = \pi_i(\mathrm{nf}_{\mathrm{OPH}}(P_\star, N_\star))$$

is uniquely determined and invariant under admissible implementation, scheduling, and hidden-carrier changes. The hard part is showing that OPH leaves no rival source map; Banach then supplies endpoint uniqueness. The theorem proves unique generation from the frozen branch. Physical identification requires the data comparison.

Read the equation literally. The simulated record world enters the simulating readback rule, contradictory branches die, and the returned object is the same record world. That is the self-contained consistency computation [O1,O2]. A universe with unexplained external constants fails this test [O1,O3,O4]. The finite OPH patch-carrier pipeline is the concrete fixed-cutoff carrier for this computation: a finite patch system with ports, central records, repair moves, checkpoints, and a terminal normal form. It carries the closure solve; frame rendering is absent.

The observer lock changes the meaning of experience. An observer in OPH is a finite record boundary with a comparison rule for nearby patches [O1,O2]. Subjective experience is that boundary reading itself from the inside:

$$r_i = \mathcal{R}_i(\mathfrak{U}_{\mathrm{OPH}}), \qquad \mathfrak{U}_{\mathrm{OPH}} = T(\{r_i\}_{\mathrm{compat}}).$$

In this equation, r_i is observer patch i's local readout, $\mathcal{R}_i$ is its readout map, and $\{r_i\}_{\mathrm{compat}}$ is the compatible readout set. The map T repairs those records and returns the shared world. Left equation: subjective experience. Right equation: shared reality. Experience matters because OPH reality is the stable object made by compatible observer readouts [O1,O2]. Private incompatible records fail to become public records. The mechanism is mundane: records read, compare, repair, return.

Exact quantum event-surface hit

The fixed-cutoff record interface also emits textbook quantum readout. On a completed compare/write/verify slice, observer-accessible events generate a finite central record algebra. For an event E,

$$\mathbb{P}(E) = \mathrm{Tr}(\rho P_E), \qquad \rho|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

These are the Born rule and the Lueders update on the OPH record surface. On the corresponding two-wing Bell event surface, the CHSH expression obeys

$$|S_{\mathrm{CHSH}}| \le 2\sqrt{2}.$$

This is an exact structural output of the finite record algebra. It uses no $\alpha(0)$, G, QCD, charged-lepton masses, or neutrino data [O1,O3,A18,A19].

2. Closure coordinates and counted witnesses

Standard physics carries roughly thirty independent numerical inputs, depending on how the count is made. OPH leaves a local pixel coordinate and a global screen-capacity coordinate. They are fixed by closure equations from the OPH rules [O1,O3,O4,O5,O6]. That is the zero-parameter claim. Observations may tell us which branch to inspect. They label the basin; closure sets the endpoint.

For every counted compact calculation, write

$$P_{\mathrm{src}} := P_{\mathrm{cand}} = 1.630972095694329\ldots.$$

This is the source-side pixel candidate emitted by the declared OPH trunk. The CODATA/NIST comparison coordinate is

$$P_C := \varphi + \frac{\sqrt{\pi}}{A_C}, \qquad A_C := 137.035999177 = 1.630968209403959\ldots.$$

The CODATA/NIST measured endpoint is a branch locator and comparison column only; it supplies no upstream source-map input. The hadron-completed OPH endpoint is $P_{\rm full}$. It becomes a counted endpoint row only when the missing QCD/hadronic electromagnetic endpoint map is emitted by the OPH rules [O5,O6]. The global screen target remains

$$N_\star \simeq N_{\rm scr}^{\rm dS} \simeq 3.31 \times 10^{122}.$$

In ordinary language, one elementary observer cell must read the same as simulated record content and simulating screen rule, and the universe must carry the record capacity required by the screen that contains that same universe.

Independent two-closure lock

The OPH universe is a self-reference. The simulating side supplies a readback rule, the simulated side supplies the record readout, and a consistent universe exists only when both sides close on the same object [O1,O2,O4]. The two closure equations are where this strange loop becomes a number. **The closure coordinates must be computed as OPH endpoints. The simulation is consistent only when the self-references hold.**

Local loop: try a pixel P_k. The OPH branch reads its field width $R_P(P_k)$. The pixel map turns that readout back into a corrected pixel:

$$\alpha_k = R_P(P_k), \qquad P_{k+1} = \Gamma_P(P_k) = \varphi + \alpha_k\sqrt{\pi}.$$

Solving $P = \Gamma_P(P)$ gives the simulating-side pixel setting for that branch. In the compact hierarchy row this coordinate is $P_{\rm src} = P_{\rm cand}$. The simulated-side field width comes from the same endpoint: $\alpha_{\rm root} = R_P(P_{\rm src})$.

Global loop: try a capacity N_k. The capacity map counts closed-screen normal forms, charges the screen budget, applies the Minimal Admissible Realization (MAR) selector, and returns the admissible capacity:

$$\Lambda_k = R_N(N_k), \qquad N_{k+1} = F_N(N_k).$$

Solving $N = F_N(N)$ gives the simulating-side capacity setting $N_\star$. The simulated-side horizon readout comes from the same computed endpoint: $\Lambda_\star = R_N(N_\star)$.

Solvers may start in observation-suggested intervals. The equations themselves must contain no measured α, measured Λ, weak masses, calibrated target residuals, or equally admissible deformations. Once the source maps are fixed and contractive, Banach gives one local endpoint and one global endpoint:

$$P_{\rm src} = \Gamma_P(P_{\rm src}), \qquad N_\star = F_N(N_\star).$$

Changing either endpoint breaks the resonance between the source setting and the universe read from inside [O1,O3,O4,O5].

The OPH screen branch supplies twelve curvature ports. Reversible write/check orientation doubles the product-adjoint repair register to 24. The local half of the lock appears as the 12-port exponent in the hierarchy proof below. This is where the fixed-point simulation stops being a metaphor. The pixel value and the screen capacity resonate: changing either one breaks the pixel-screen closure and removes the consistent universe. Measurements can label the basin; closure supplies the decimal.

The same endpoints have a limited carrier reading. $P_{\rm src}$ fixes a canonical geometric cell area, and $N_\star$ fixes total screen capacity. Under an equal-area screen chart the cell count is $K_{\rm cell} = 4N_\star/P_{\rm src}$. This leaves the primitive observer algebra open. The quantity $P_{\rm src}/4$ is shared cut entropy density, so the carrier reading remains a variable finite federation unless a separate theorem selects an isomorphic one-cell or fixed-block primitive carrier.

The local constant is computed from a readback equation with measured α excluded [O3,O4,O5,O6]:

$$P \longmapsto \alpha_U(P) \longmapsto A_Z(P) \longmapsto A_T^{\rm src}(P), \qquad \Gamma_{\rm src}(P) = \varphi + \frac{\sqrt{\pi}}{A_T^{\rm src}(P)}.$$

Read left to right: try a pixel value P, let the OPH branch emit its field widths, and let $\Gamma_{\rm src}$ return the corrected pixel value. The symbol P is a candidate identity value for one OPH pixel: the number that converts the cell's screen area into the edge entropy it shares with neighboring cells. The symbols $\alpha_U(P)$, $A_Z(P)$, and $A_T^{\rm src}(P)$ are the three OPH readouts along the way: finite-screen unified gauge width, electroweak anchor, and Ward-projected Thomson endpoint.

The public implementation solves the same loop in the α-chart [C1]. Start with a trial α_k. It defines a candidate pixel

$$P_k = \varphi + \alpha_k\sqrt{\pi}.$$

From P_k, the OPH branch computes the unification point, the electroweak anchor, and the Thomson endpoint. That returns the inner electromagnetic readout

$$\alpha_{\rm in}(P_k) = \frac{1}{A_T^{\rm src}(P_k)}.$$

The solver asks how far the returned α is from the trial α:

$$\rho(\alpha_k) = \alpha_{\rm in}(\varphi + \alpha_k\sqrt{\pi}) - \alpha_k.$$

The code scans the declared branch interval, finds where ρ crosses zero, and cuts the bracket in half until the mismatch is below the requested tolerance. In fixed-point notation, the same loop is

$$\alpha_{k+1} = \alpha_{\rm in}(\varphi + \alpha_k\sqrt{\pi}), \qquad P_{k+1} = \varphi + \alpha_{k+1}\sqrt{\pi}.$$

The compact hierarchy output is the pair

$$P_{\rm src} = P_{\rm cand}, \qquad \alpha_{\rm root}^{-1} = 136.9948351646\ldots.$$

CODATA never enters this map; it can only be printed after the solve as a comparison.

The simulating geometric reading and simulated electromagnetic reading are

$$\alpha_{\rm ext}(P) = \frac{P-\varphi}{\sqrt{\pi}}, \qquad \alpha_{\rm in}(P) = \frac{1}{A_T^{\rm src}(P)}, \qquad \alpha_{\rm ext}(P_{\rm src}) = \alpha_{\rm in}(P_{\rm src}),$$

The last equality is the pixel lock, equivalently $P_{\rm src} = \Gamma_{\rm src}(P_{\rm src})$. The golden ratio $\varphi = (1+\sqrt{5})/2$ is the geometric screen balance point. A purely golden pixel has geometry with no field-width readback. $\sqrt{\pi}$ converts boundary width into the pixel chart, and the electromagnetic width supplies the detuning. The low-energy electromagnetic width is read out as

$$\alpha_{\rm root} = \frac{1}{A_T^{\rm src}(P_{\rm src})} = \frac{P_{\rm src} - \varphi}{\sqrt{\pi}}.$$

Solving the pixel equation determines $P_{\rm src}$ and the branch field width together. Approximate measured α may identify the branch interval; it never defines $A_T^{\rm src}$ or $\Gamma_{\rm src}$. At the fixed point one may write

$$P_{\rm src} = \varphi + \alpha_{\rm root}\sqrt{\pi}$$

as the readout identity of the solved loop.

Measured α is only a comparison

The pixel equation has two readings of the same cell: the outer screen width $\alpha_{\rm ext}(P) = (P-\varphi)/\sqrt{\pi}$ and the inner electromagnetic reading $\alpha_{\rm in}(P) = 1/A_T^{\rm src}(P)$. The fixed point is their shared value:

$$P_{\rm src} = \Gamma_{\rm src}(P_{\rm src}), \qquad \alpha_{\rm root} = \alpha_{\rm ext}(P_{\rm src}) = \alpha_{\rm in}(P_{\rm src}).$$

Reading the identity backward with CODATA/NIST $A_{\rm C} = \alpha^{-1}(0)$ gives the comparison coordinate $P_C = \varphi + \sqrt{\pi}/A_{\rm C}$. The counted OPH calculation is the closure solve. The code path in [C1] keeps comparison values out of the solver.

Fine-structure prediction and QCD endpoint

The fine-structure row is the sharp numerical stress test in this compact proof. The source map solves the OPH pixel loop with no measured $\alpha(0)$, no Thomson endpoint, and no hadron data in the input path. The trunk emits $\alpha_{\rm root}^{-1} = 136.9948351646\ldots$, and the same branch emits $\alpha_U(P_\star) = 0.041124336195630495$. Together they give the source-side no-hadron near-endpoint

$$A^{\rm fp}_{\alpha_U} = \alpha_{\rm root}^{-1} + \alpha_U(P_\star) = 137.035959500817279952949994585787\ldots,$$

compared with the CODATA/NIST measured Thomson endpoint $A_{\rm C} = \alpha^{-1}(0) = 137.035999177(21)$. The remaining CODATA/NIST comparison gap is

$$0.000039676182720047050005414213\ldots,$$

only 2.8953×10^{-7} relative, or 0.000028953%. This is a one-part-in-3.45×10^6 source-side no-hadron near-endpoint prediction. The residual sits exactly where the compact proof says the source trunk stops: low-energy electromagnetic dressing by confined quark and hadron spectral transport.

In the fine-structure notation, the post-α_U missing correction is

$$\Delta^{\rm fp}_{{\rm H},req} = A_{\rm C} - A^{\rm fp}_{\alpha_U} = \alpha_U(P_\star)\,(C_{24,Q} - 1),$$

with

$$C_{24,Q} = 1.000964785973232625384951114070 2475\ldots.$$

The full root-to-CODATA difference is only bookkeeping:

$$\Delta^{\rm fp}_{{\rm H},cal} = A_{\rm C} - \alpha_{\rm root}^{-1} = 0.041164012378350542050005414213\ldots = \alpha_U(P_\star)C_{24,Q}.$$

The factor $C_{24,Q}$ names the same-scheme low-energy hadronic endpoint transport required for source-only closure. A 24-register spectral object fit after looking at the Thomson residual is exact accounting rather than prediction.

The parameter-free trunk value is reproducible from the public solver and checked runtime report [C1]:

```
cd code/P_derivation && python3 derive_p.py --no-hadron-closure
```

The saved report `runtime/full_p_alpha_report_current.json` prints $\alpha^{-1} = 136.994835164621649457949994585787\ldots$. The measured CODATA value never enters that command path [C1].

The open piece is the low-energy QCD/hadronic correction: confined quark and hadron effects in the electromagnetic endpoint. OPH source-only closure requires the Ward-projected hadronic spectral measure and same-scheme finite endpoint remainder without Thomson-target leakage. That finite-lattice QCD calculation is outside this compact proof; OMEGA is the optical fixed-point compute lane for such payloads [S14]. CODATA/NIST $\alpha(0)$ therefore stays in the calibrated comparison lane [O5,O6].

The compact proof keeps the CODATA/NIST $\alpha(0)$ endpoint, hadron masses, and SI G clock row outside its core count. It uses the QCD-free hierarchy witness below, which avoids the hadronic correction completely. $P_{\rm full}$ names the hadron-completed endpoint defined by that OPH QCD map.

The global constant is computed from the screen-capacity readback equation [O1,O4]:

$$N_\star = \mathcal{C}(N_\star).$$

The universe is allowed exactly the record capacity that its own closed-screen readback says it needs. The symbol N is a candidate total capacity for the closed observer screen: the number of record units the universe can carry in OPH units. $\mathcal{C}$ is the closed-screen readback rule. The simulated reading counts the closed-screen normal forms supported at capacity N. The simulating reading charges the screen for the N slots it claims. A self-contained fixed-point computation has no external memory bank, so the available records and the cost of the screen must close on the same capacity. The global loop says the simulated universe contains the capacity of the simulating screen that contains it [O1,O4]. The screen solver mirrors the pixel solver. Try a capacity N, count the closed-screen normal forms it supports, subtract the screen budget charge, and apply MAR to pick the admissible branch. The score is

$$S(N) = \log|\Omega_N^{\rm sc}| - N, \qquad N_\star = \mathrm{MAR}\arg\max_N S(N) \simeq 3.31 \times 10^{122}.$$

The set $\Omega_N^{\rm sc}$ contains the closed-screen normal forms at capacity N. The term $\log|\Omega_N^{\rm sc}|$ is the simulated-side count, and the subtraction of N is the simulating-side budget charge. The operator MAR is Axiom 5 of OPH [O1]: Minimal Admissible Realization. In the core papers, Axiom 5 says that the realized branch is the lexicographically minimal admissible representative under the branch complexity vector $C(\mathfrak{S})$. In this screen-capacity display, the same selector chooses the minimal admissible arithmetic representative of the maximizing branch. The output is the unique selected capacity. The cosmological constant comes from that same capacity:

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \qquad \Lambda_\star a_{\rm cell} = \frac{3\pi P_{\rm src}}{N_\star}.$$

So $N_\star$ and $\Lambda_\star$ are computed together in the global closure, the same way $P_{\rm src}$ and $\alpha_{\rm root}$ are computed together in the local closure [O1,O4].

The three-scalar cross-lock

Before any decimal appears, the recovered compact-gauge branch fixes the Standard Model quotient

$$G_{\rm SM} = \frac{\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3.$$

The derivation has a specific route. Overlap holonomy and zero-obstruction transport first build a compact bosonic gauge category. DR/Tannaka reconstruction reads off a compact group from that category. MAR then selects the realized one-generation/one-Higgs matter package; anomaly cancellation, Yukawa invariance, and the central kernel of the realized action fix the hypercharge lattice and the $\mathbb{Z}_6$ quotient [O4,A21]. The same branch carries the protected massless photon, gluon, and graviton rows:

$$m_\gamma = 0, \qquad m_g = 0, \qquad m_{\rm grav} = 0.$$

The same quotient fixes charge quantization: $Q = T_3 + Y$ and $Q_{\text{color singlet}} \in \mathbb{Z}$. Its connected adjoint is $(8,1,0)\oplus(1,3,0)\oplus(1,1,0)$. There are no mixed $(3,2,\pm5/6)$ X/Y gauge bosons, so the ordinary simple-GUT gauge-mediated proton-decay channel is absent on this branch [O4,A20,A21]. This discrete structural witness sits upstream of the numerical rows. The same branch then emits the compact integer triple

$$N_c = 3, \qquad N_g = 3, \qquad \beta_{\rm EW} = N_c + 1 = 4.$$

These three integers feed both compact numerical witnesses below. The same 4 controls the charged-family phase and the weak hierarchy exponent [O3,O4,O5]. One source branch gives one huge hierarchy and a two-coordinate lepton-family shape.

Phase-rigidity charged-family prediction. The charged-family row is one correlated family-shape prediction: no QCD, no G, no public $\alpha(0)$, and no charged-lepton masses enter as inputs. On the realized Standard Model branch,

$$N_c = 3, \qquad N_g = 3, \qquad \beta_{\rm EW} = N_c + 1 = 4.$$

The OPH-specific claim is the phase selector. A balanced three-family carrier sits on the Koide cone for any phase. OPH fixes the phase from branch integers:

$$\delta_e = \frac{\beta_{\rm EW}}{2N_cN_g} = \frac{2}{9}.$$

If R is the cyclic three-family shift, $R^3 = I$, define

$$C_{\delta_e} = I + \frac{1}{\sqrt{2}}\left(e^{i\delta_e}R + e^{-i\delta_e}R^2\right).$$

Its eigenvalues are

$$r_k = 1 + \sqrt{2}\cos\left(\delta_e + \frac{2\pi k}{3}\right), \qquad k = 0,1,2.$$

For $\delta_e = 2/9$, the ordered roots are

$$(r_e, r_\mu, r_\tau) = (0.040349908219207475,\ 0.5802119201475368,\ 2.3794381716332555).$$

The trace identities are immediate:

$$\operatorname{tr} C_{\delta_e} = 3, \qquad \operatorname{tr} C_{\delta_e}^2 = 6.$$

If charged square-root masses lie on this OPH carrier direction, $m_i = s_e r_i^2$, then Koide follows exactly:

$$\frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{s_e \operatorname{tr} C_{\delta_e}^2}{s_e (\operatorname{tr} C_{\delta_e})^2} = \frac{6}{9} = \frac{2}{3}.$$

The phase-fixed direction predicts the two ordered scale-free charged-family ratios

$$\frac{m_\mu}{m_e} = \left(\frac{r_\mu}{r_e}\right)^2 = 206.770315972729\ldots,$$
$$\frac{m_\tau}{m_e} = \left(\frac{r_\tau}{r_e}\right)^2 = 3477.47283710460\ldots,$$
$$\frac{m_\tau}{m_\mu} = \left(\frac{r_\tau}{r_\mu}\right)^2 = 16.8180467333776\ldots.$$

Using the PDG lepton summary masses $m_e = 0.51099895000\,\mathrm{MeV}$, $m_\mu = 105.6583755\,\mathrm{MeV}$, and $m_\tau = 1776.86\,\mathrm{MeV}$, the deviations are about 9.8 ppm, 70 ppm, and 60 ppm respectively [S8]. The Koide value computed from those same pole masses is about 9.2 ppm below 2/3. The compact row counts one phase-rigidity prediction: $\delta_e = 2/9$ fixes the normalized charged-family direction, and the two ordered ratios test that direction jointly. This row fails if N_c, N_g, or $\beta_{\rm EW}$ lack an independent Standard Model-branch derivation, if $\delta_e = 2/9$ is chosen from charged-lepton masses, or if the charged-family shape misses the sorted lepton direction [O5,O6,A15,A16].

The QCD-free hierarchy witness

The hierarchy problem becomes a QCD-free branch-output calculation. The second independent numerical witness from the same structural branch stays away from QCD. A route through the public Thomson endpoint, hadron masses, or the cesium clock would import the Ward-projected hadronic correction as an assumption. That criticism is fair, so this row avoids it. The OPH calculation fixes $\alpha_U(P_{\rm src}) = 0.041124336195630495$ in the interval $I_U = [0.041123336195630494, 0.041125336195630496]$, with the Krawczyk image strictly inside I_U [O5]. The counted row is the weak-to-UV hierarchy emitted from $P_{\rm src}$ and that independently checked $\alpha_U(P_{\rm src})$ value [O5,C3]. OPH gets the tiny weak scale from the two closure constants, without importing hadronic physics.

The factor 12 is screen topology doing the counting. On the triangulated S^2 screen, the locally flat triangular vacuum is six-valent. Define the curvature charge $q_v = 6 - \deg(v)$. Since $V - E + F = 2$ and $3F = 2E$, one has $E = 3V - 6$, hence

$$\sum_v q_v = 6V - \sum_v \deg(v) = 6V - 2E = 12.$$

A closed triangular screen must carry exactly twelve units of curvature. Refinement-stable MaxEnt spreads this total charge into twelve unit fivefold defects: at fixed $\sum q_v = 12$, the convex local defect cost $\sum q_v^2$ is minimized by unit charges, while negative defects require extra positive charge. Edge-center completion turns each defect collar into a central port [O3]. With no marked point, the twelve equal ports sit on the maximal finite rotation orbit A_5/C_5, whose size is $60/5 = 12$. Therefore the global screen depth $D = \log(N/\pi)$ is sampled locally as $D/12$.

The representation-to-spectrum step turns the same screen count into the oriented repair register:

$$m_{\rm rep} = 2\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8+3+1) = 24.$$

The factor two is reversible write/check orientation. Since $N = \pi R_{\rm dS}^2/\ell_\star^2$, the same $D = \log(N/\pi)$ is the cosmic length ratio in logarithmic form. The electroweak branch samples that global depth through the 12 unoriented ports and the $4P_{\rm src}$ projection below; the 24-slot register supplies the repair normalization [O4,O5,C3].

The local cell entropy is $\ell_{\rm cell} = P_{\rm src}/4$, and the electroweak projection coefficient is $\beta_{\rm EW} = N_c + 1 = 4$. Therefore

$$\Gamma_{\rm EW} = \beta_{\rm EW}\ell_{\rm cell}\frac{\log(N/\pi)}{12} = \frac{P_{\rm src}}{12}\log(N/\pi).$$

This is the local share of the global screen step: cell entropy, times the electroweak projection, times one twelfth of the screen depth. The compact standalone hierarchy witness is the transmutation law

$$\frac{v}{E_\star} = P_{\rm src}^{-1/2}\exp\left[-\frac{2\pi}{4\alpha_U(P_{\rm src})}\right] = 2.0199803239725553 \times 10^{-17}.$$

$E_\star$ is the OPH UV energy scale. $E_{\rm cell} = E_\star/\sqrt{P_{\rm src}}$ is the cell energy after the pixel area is accounted for. The exponent uses the OPH-emitted $\alpha_U(P_{\rm src})$. It excludes G, Λ, M_W, M_Z, Higgs mass, the public Thomson endpoint, and QCD hadronic input [O5]. This is the QCD-free quantitative witness.

The bridge coordinate rewrites the same exponent as a screen-capacity readout:

$$\frac{v}{E_{\rm cell}} = \left(\frac{N_{\rm bridge}^{\rm EW}}{\pi}\right)^{-P_{\rm src}/12}, \qquad N_{\rm bridge}^{\rm EW} = \pi\exp\left[\frac{6\pi}{P_{\rm src}\alpha_U(P_{\rm src})}\right].$$

$N_{\rm bridge}^{\rm EW}$ is a derived capacity readout on the local/global hierarchy branch. It is bookkeeping only; empirical closure comes from the source capacity row. The affine map

$$\mathcal{C}_{\rm EW}(P,x) = (1-\lambda)x + \lambda\frac{6\pi}{P\alpha_U(P)}, \qquad \lambda = \frac{1}{2}.$$

For fixed $P = P_{\rm src}$, this is a contraction with unique endpoint

$$x_\star = \frac{6\pi}{P_{\rm src}\alpha_U(P_{\rm src})}, \qquad N_{\rm CRC}^{\rm EW} = \pi e^{x_\star} = 3.5323546226929906\ldots \times 10^{122}.$$

The residual

$$R_{\rm EW} := \alpha_U(P_{\rm src})^{-1} - \frac{P_{\rm src}}{6\pi}\log(N_{\rm CRC}^{\rm EW}/\pi)$$

vanishes at that endpoint by definition, equivalently

$$\alpha_U(P_{\rm src})\log(N_{\rm CRC}^{\rm EW}/\pi) - \frac{6\pi}{P_{\rm src}} = 0.$$

The finite readback-resolution result supplies the positive-root readback limit

$$\rho_{\rm read}(r, N_{\rm CRC}^{\rm EW}) \longrightarrow (N_{\rm CRC}^{\rm EW}/\pi)^{-1/2}.$$

Together with the 12-port screen sieve and the 24-slot oriented repair register, this gives an identity form of the hierarchy witness:

$$t_{\rm tr}(P_{\rm src}) = \frac{P_{\rm src}}{12}\log(N_{\rm CRC}^{\rm EW}/\pi), \qquad \frac{v}{E_{\rm cell}} = (N_{\rm CRC}^{\rm EW}/\pi)^{-P_{\rm src}/12}.$$

The compact count is therefore one computed hierarchy closure and one global capacity target. A source-only global capacity row carries empirical weight only after an independently motivated executable map computes screen capacity without measured Λ, horizon radius, age, or $N_{\rm scr}$, and then compares the frozen output with cosmology [O4,O5,C3].

Ordinary particle physics treats the Higgs scale as a tuning problem. In OPH the weak scale is a branch output with zero naturality defect:

$$\epsilon_H = 0, \qquad \epsilon_H \in [0,0].$$

Zero means there is no cancellation to arrange. There is no bare scalar dial to balance against a cutoff; the weak scale is the settled-form readout of the 24-slot oriented repair-register normalization. That is the hierarchy problem in standard language, with the standard fine-tuning term absent from the OPH normal form [O4,O5].

Discrete rigidity check

The integers $N_c = 3$, $N_g = 3$, and $\beta_{\mathrm{EW}} = 4$ are branch data with no extra measured decimals. The evidence is the overconstraint. Once the Standard Model branch fixes this integer triple, the same β_{EW} hits two unrelated readouts: the weak hierarchy and the charged-family shape. Hold the P_{src} and $\alpha_U(P_{\mathrm{src}})$ proof records fixed and change only β_{EW}:

β_{EW}	$v/E_\star$	m_μ/m_e	m_τ/m_e
3	5.97×10^{-23}	25.84	579.07
4	2.01998×10^{-17}	206.7703	3477.47
5	4.20×10^{-14}	outside cone	outside cone

The $\beta_{\mathrm{EW}} = 5$ charged carrier has

$$r_e = 1 + \sqrt{2}\cos\left(\frac{5}{18} + \frac{4\pi}{3}\right) = -0.0158500583\ldots,$$

so the light square-root mass leaves the positive cone. The nearby branch values miss both sectors at once. The 4 is therefore a positive-edge branch: it keeps the hierarchy in the observed sector and produces a tiny positive electron root without fitting charged masses [O4,O5,C3].

Prediction escrow: rows exposed to failure

Known constants can point to branch neighborhoods. The case needs frozen outputs outside the charged-family and hierarchy rows, exposed to external measurements.

Neutrino weighted-cycle branch. The weighted-cycle branch emits

$$(m_1, m_2, m_3) = (0.017454720257976796,\ 0.019481987935919015,\ 0.05307522145074924)\ \mathrm{eV},$$

with

$$\sum m_\nu = 0.09001192964464505\ \mathrm{eV}, \quad \Delta m^2_{21} = 7.488059465106851 \times 10^{-5}\ \mathrm{eV}^2,$$

$$\Delta m^2_{31} = 2.5123118727618473 \times 10^{-3}\ \mathrm{eV}^2, \qquad (\alpha_{21}, \alpha_{31}) = (153.618518^\circ, 257.003241^\circ).$$

The same branch emits the PMNS readout

$$(\theta_{12}, \theta_{23}, \theta_{13}) = (34.225904631810025^\circ,\ 49.72282845058266^\circ,\ 8.686355527700156^\circ),$$

$$\delta_{\mathrm{PMNS}} = 305.58061231449796^\circ, \qquad J = -0.02753115613565372, \qquad \frac{\Delta m^2_{21}}{\Delta m^2_{32}} = 0.030721110097966534.$$

The same output gives direct laboratory and cosmology targets. With $c_{ij} = \cos\theta_{ij}$ and $s_{ij} = \sin\theta_{ij}$,

$$m_\beta = \sqrt{c^2_{12}c^2_{13}m^2_1 + s^2_{12}c^2_{13}m^2_2 + s^2_{13}m^2_3} = 0.0196244372655102\ \mathrm{eV}.$$

For neutrinoless double beta decay, set $\alpha^{0\nu}_{31} = \alpha_{31} - 2\delta_{\mathrm{PMNS}}$. The branch gives

$$m_{\beta\beta} = \left| c^2_{12}c^2_{13}m_1 + s^2_{12}c^2_{13}m_2 e^{i\alpha_{21}} + s^2_{13}m_3 e^{i\alpha^{0\nu}_{31}} \right| = 0.00797676286284509\ \mathrm{eV}.$$

This row is independent of the hierarchy and charged-family shape. Oscillation, absolute-mass, cosmology, beta-endpoint, or neutrinoless-double-beta data can reject the branch. In the particle paper and code it is theorem-level on the declared weighted-cycle branch; the positive-segment adapter and bridge-coordinate readouts stay diagnostic and feed nothing back into the theorem state [O5,C2,A17,S9,S10,S11].

Charged-family residual. The known Koide relation is background. The OPH claim is the branch phase and the residual. Define

$$\eta_e := \delta_e^{\mathrm{phys}} - \frac{2}{9}.$$

The compact row uses $\delta_e = 2/9$ as a branch cross-lock with the hierarchy integer triple. A full charged-family promotion has to emit η_e and the affine determinant-line anchor from the same OPH branch [O5,O6,A15,A16].

Running-scheme rigidity. The hierarchy row counts only when the $\alpha_U(P_{\mathrm{src}})$ proof record fixes the high-scale beta packet, threshold packet, representation cutoffs, matching prescription, and renormalization convention outside the target value. Tuning any of those with $v/E_\star$ as the target demotes the row to calibration. A stronger closure emits the running packet from the same OPH branch, or emits a threshold spectrum that experiments can exclude [O5,C3,A8,A10].

Row	Role in this note	Decisive check
$v/E_\star$	Measurement-free hierarchy witness	Frozen $\alpha_U(P_{\mathrm{src}})$ packet with no measured-answer path.
m_μ/m_e, m_τ/m_e	Scale-free charged-family shape	OPH map for η_e and the affine determinant-line anchor.
ν masses, phases, and endpoint targets	Prediction escrow	External data accept or exclude the frozen branch.
$\alpha(0)$	Empirical endpoint closure	$\alpha^{-1}_{\mathrm{root}} = 136.9948351646\ldots$, and $A^{\mathrm{fp}}_{\alpha_U} = 137.0359595008\ldots$; OPH hadronic spectral map must emit the remaining QCD correction without target leakage.
G_{SI}	Clock-hierarchy display	No-G cesium gap is emitted without a gravity-calibrated proxy.

3. Geometric gravity normalization

Einstein theorem boundary
Bare finite consensus supplies quotient normal forms; Einstein dynamics belongs to the declared branch $\mathsf{Cons}_r + \mathsf{GeomRead}_r + \mathsf{BW}_r + \mathsf{NullStress}_r + \mathsf{BoundedInterval}_r + \mathsf{FixedCapStat}_r + \mathsf{SmallBallArea}_r + \mathsf{TensorUpgrade}_r$, with D6 capacity closure for Λ. Failure of any branch clause leaves the gravity row unclosed.

OPH gets gravity before meters, kilograms, or seconds enter the discussion. G_{geom} is Newton coupling in OPH geometric units, a_{cell} is one observer-cell area, $\bar{\ell}_{\text{shared}}$ is its shared edge-entropy weight, and $\ell_\star$ is the OPH length quantum. The edge-entropy/area-law theorem on the OPH gravity branch identifies $a_{\text{cell}}/(4\bar{\ell}_{\text{shared}})$ with the structural coefficient of the stress-energy term in the local Einstein equation $G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}}\langle T_{ab}\rangle$ and in the Newton-Poisson weak-field limit $\nabla^2\Phi = 4\pi G_{\text{geom}}\rho$; the same G_{geom} propagates literally through the Jacobson chain into both limits [O4,A22]. On the realized local-pixel branch that theorem reduces to

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}}, \qquad a_{\text{cell}} = P_{\text{src}}\ell_\star^2, \qquad \bar{\ell}_{\text{shared}} = \frac{P_{\text{src}}}{4}.$$

The A22 derivation fixes the factor 4. Substitution gives

$$\frac{G_{\text{geom}}}{\ell_\star^2} = \frac{P_{\text{src}}}{4(P_{\text{src}}/4)} = 1, \qquad G_{\text{geom}} = \ell_\star^2.$$

The same pixel number counts area and shared entropy. The pixel number cancels. Gravity is the area quantum in OPH units. The SI decimal needs the separate no-G clock proof [O4,A6]. The edge-entropy/area-law theorem itself uses no measured G, no Planck area, no measured Λ, and no gravity-calibrated clock scale [O4,A22]. In the same theorem, the linear-in-area scaling $\langle L_C\rangle \propto A(\partial C)$ is derived inside OPH from the Markov-collar/refinement-stability structure of the screen net, the OPH structural Einstein-equation coefficient is then $\kappa_{\text{OPH}} = 2\pi\, a_{\text{cell}}/\bar{\ell}_{\text{shared}}$, and the Newton-Poisson convention $\nabla^2\Phi = 4\pi G_N\rho$ defining the Newton coupling fixes $G_N = \kappa_{\text{OPH}}/(8\pi) = a_{\text{cell}}/(4\bar{\ell}_{\text{shared}}) = G_{\text{geom}}$. The factor of 4 is the ratio $\kappa_{\text{OPH}}/(8\pi)$.

4. Compact proof rows

The compact argument separates three evidence classes. Internal theorems show mathematical coherence of the declared branch. Retrodictive matches compare source-only outputs with known quantities. Prospective predictions are frozen outputs exposed to later tests. The compact proof surface consists of: observer-patch architecture, the exact quantum event surface, the Standard Model structural branch, the phase-fixed charged-family shape, the QCD-free hierarchy witness, and the geometric gravity normalization identity. OPH claims to describe the whole machine, so the longer particle, cosmology, dark-sector, hardware, and SI rows matter. They stay outside this compact surface unless their own calculations pass the same rule [O1,O4,O5,C2,C3].

Rows outside the compact count remain useful evidence and falsification surfaces with their own claim classes. The readable falsifiability map is `extra/OPH_falsifiability.md`.

Witness	Evidence class	Why	Refs
Finite observer-patch normal form and patch-carrier pipeline	Internal theorem	Finite observer records, overlaps, mismatch scores, repair maps, checkpoints, and patch carriers define the fixed-point architecture. The row tests the mechanism before any decimal fit enters.	[O1,O2,C5]
Born-Lueders and Tsirelson event surface	Internal theorem	The finite central record algebra gives $\mathbb{P}(E) = \mathrm{Tr}(\rho P_E)$, Lueders conditioning, and $\lvert S_{\text{CHSH}}\rvert \le 2\sqrt{2}$ on the declared Bell event surface.	[O1,O3,A18,A19]
Standard Model structural branch	Internal theorem	$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y/\mathbb{Z}_6$, $N_c = 3$, $N_g = 3$, protected massless carriers, charge quantization, and no X/Y gauge bosons are discrete structural outputs. The same integers feed the hierarchy and charged-family witnesses.	[O3,O4,O5,A20,A21]
$\alpha_U(P_{\text{src}})$ interval check	Internal theorem	The unified-coupling row is checked by an interval/Krawczyk calculation with measured electroweak masses, low-energy couplings, G, Planck units, and Λ excluded.	[O5,C2,A8]
Phase-fixed charged-family shape	Retrodictive match	$\delta_e = (N_c+1)/(2N_cN_g) = 2/9$ fixes the scale-free charged-family direction. Koide is an identity of the carrier; the two ordered lepton ratios test the single normalized direction. Absolute lepton masses stay outside this row.	[O5,O6,A15,A16]
QCD-free hierarchy law	Retrodictive match	$v/E_\star = P_{\text{src}}^{-1/2}\exp[-2\pi/(4\alpha_U(P_{\text{src}}))]$. The weak scale is emitted without the QCD/hadronic endpoint and $\epsilon_H = 0$, conditional on the independent meaning of $E_\star$.	[O5,C3,A10]
Geometric gravity normalization	Internal identity	$G_{\text{geom}} = \ell_\star^2$. This fixes the OPH area normalization. The SI Newton decimal needs the separate clock row.	[O3,O4,A7]
CODATA/NIST $\alpha(0)$ and P_C	Comparison only	The measured endpoint requires the missing QCD/hadronic electromagnetic transport. $P_C = \varphi + \sqrt{\pi}/A_C$ is a comparison coordinate under the declared missing-map condition.	[O5,O6,A13]
SI G	Comparison only	The SI decimal counts only when the no-G cesium clock hierarchy emits ε_{Cs} without measured G or gravity-calibrated proxies.	[O4,A6]
Particle, dark-sector, neutrino, cosmology, and collider rows	Prospective prediction	These rows remain useful tests with their declared claim classes. The compact anti-circularity argument keeps them outside its count.	[O5,O7,O8,C2,C4]
PIXEL, IBM, OMEGA, and hardware lanes	Experimental surface	Hardware lanes test named lemmas. The universe claim depends on the source rows and data comparison. They count only with public task definitions, raw readouts, controls, manifests, and verifier logs.	[S12,S13,S14]

The compact rows in this table are derived from OPH closure coordinates and the OPH axioms, with the full derivation chains and proofs contained in the referenced papers. Comparison and test rows keep their declared evidence class and failure conditions. For details, see the linked papers and the public code in `github.com/FloatingPragma/observer-patch-holography`.

5. Falsifiers and evidence classes

OPH has many ways to be wrong: source leakage, nonunique closure, a MAR rival, carrier-dependent readout, hidden-scale bridge, or missed frozen prediction. One failure demotes its row; a central failure kills the compact claim. The empirical test is a frozen compression score. Freeze the theory, source maps, branch selectors, tolerances, uncertainty model, comparison theories, and held-out dataset D_{test}. Then compare

$$\Delta_{\text{bits}} = \log_2 \frac{p(D_{\text{test}} \mid M_{\text{OPH}})}{p(D_{\text{test}} \mid M_{\text{ref}})} - \big[L(M_{\text{OPH}}) - L(M_{\text{ref}})\big].$$

The length $L(M)$ charges the full description of the model: axioms, MAR, source equations, branch rules, selectors, tolerances, and search. The empirical identification claim is that a source-locked OPH generation compresses the preregistered held-out observations by at least the declared threshold Δ_0 bits more than the reference class.

Physics problems that become easy to solve with OPH

One fitted constant is cheap. This claim has a harder job. It has to hit the compact rows and make the observer problem, gravity/gauge reconstruction, hierarchy, dark energy, dark matter, Yang-Mills gap, unification without proton decay, particle inventory, and string vacuum selection fall out of the same two fixed points. String theory earned attention because its quantized strings naturally contain a spin-2 graviton. OPH produces the protected massless spin-2 branch in the observer-overlap reconstruction, and the string-vacuum paper shows why string theory is an effective description of the same controlled compact branch [O4,O9,S15].

Observer problem: standard physics has no clean public-world role for subjective experience. OPH makes experience a finite readout boundary, and public reality is the compatible fixed point of those readouts [O1,O2].

Gravity with gauge physics: standard physics keeps general relativity and gauge theory in separate formalisms. OPH puts the Einstein-branch geometry readout, Lorentz kinematics, the $H^3 \simeq \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$ three-dimensional spatial chart, the Jacobson-Einstein branch, compact gauge reconstruction, the Standard Model quotient, hypercharge, three colors, and three generations in one observer-overlap reconstruction stack [O3,O4,O5].

Hierarchy problem: OPH turns hard Higgs naturality into a branch-output calculation. The 12-port screen sieve, the 24-slot oriented repair register, and the selected local/global capacity bridge emit the weak scale with $\epsilon_H = 0$, so the usual fine-tuning term disappears [O4,O5,C3].

Yang-Mills gap: OPH turns the Clay-style gap problem into a repair gap statement. On a controlled compact-gauge branch, after the continuum proof in [O10] is supplied, the Yang-Mills mass gap is $\Delta_{\text{YM}} = \Delta_{\text{rep}} > 0$. Without that proof, finite repair remains regulator-level gap accounting [O4,O10].

Dark energy: OPH turns the cosmological-constant problem into global screen bookkeeping. The cosmological constant is the closed-screen capacity readout of the same Einstein branch. Arbitrary vacuum-energy input has no role in that row [O1,O4].

Dark matter: OPH replaces an unseen-particle assumption by repair-stress bookkeeping on its declared dark branch. The public dark-sector rows have continuation and scorecard status; full likelihood and calculation chain closure is their separate test. The static galaxy continuation gives the functional target $\nu_{\text{OPH}}(x) = [1 - \exp(-\lambda_{\text{collar}}\sqrt{x})]^{-1}$, with $g_{\text{obs}} \simeq \sqrt{a_{\text{eff}} g_b}$ and $v^4 = GM_b a_{\text{eff}}$ in the deep low-acceleration limit [O7,O8,C4].

Unification without proton decay: OPH gets unification-style running without the classic simple-GUT proton-decay problem. The edge branch runs unification-style while the realized product quotient has no simple-GUT leptoquark generators [O4,O5].

Particle inventory: the compact proof counts structural carriers, gauge quotient data, the phase-fixed charged-family shape witness, the $\alpha_U(P)$ interval check, and the QCD-free hierarchy witness. W/Z, absolute charged-lepton masses, selected-frame quarks, Higgs/top, and neutrino rows remain in the particle files with their declared support levels [O4,O5,C2].

String vacuum selection: the same observer-patch constraints define a vacuum sieve over critical-string candidates. The string-vacuum note reads strings as an effective edge language and selects the Bouchard-Donagi witness only behind the critical-edge, cohomology, safety-layer, threshold, and moduli-locking gates. This is the precise sense in which OPH recovers string theory as an effective description inside the OPH stack; no ordinary string vacuum is promoted to the OPH-native vacuum without those checks [O9,S15].

Tests outside the compact proof count

The next entries are clean ways to break OPH branches. They sit outside the compact proof until their own calculation and checks are supplied without reading the measured answer.

Conditional screen-spectrum and CMB lift [O2,O4]. The compact/main paper states the finite-screen theorem as a conditional continuation: scalar readout and inverse-repair receipts can promote the branch to a MaxEnt Gaussian screen covariance and the

$$\theta_{\mathrm{scr}} = P/48$$

screen exponent. Primordial curvature amplitude and TT/TE/EE spectra require the complete bridge. The complete bridge must pass before the thin-shell power-law branch write $n_s = 1 - \theta_{\mathrm{scr}}$. The source-side bridge is

$$q_\zeta = \frac{1}{3}\log\frac{J_X}{\bar{J}_X} = \zeta_\rho, \qquad \mathcal{L}_u\zeta_a = -\frac{\Theta}{3(\rho+p_{\mathrm{eff}})}\Gamma_a^{\mathrm{eff}},$$

with

$$\Gamma_a^{\mathrm{eff}} = 0 + \text{ repair gap} + \text{ rank-one mode} \Longrightarrow \text{primordial curvature}$$

only after the radial prior, null-space, phase-coherence, isocurvature, and forward-residual receipts pass. The missing steps are explicit gates: scalar fixed expected quadratic release energy, radial screen-to-primordial lift, a finite covariant collar-packet parent for any dark/anomaly source, Boltzmann transfer, source-provenance, pooled global reducers, and frozen likelihood execution. The promoted quantities $\eta_R, A_\zeta, q_{\mathrm{IR}}, \ell_{\mathrm{IR}}, B_A, \rho_A$, any checked Γ_{rec}, and N_{CRC} must come from an immutable source dependency DAG; contradictory no-data-use metadata fails closed, shard-local nonlinear averages are diagnostics, and N_{CRC} is the consensus invariant unless a disjoint additive capacity schema is supplied. A scalar $\rho_A(a), B_A(k,a)$ table plus a transition spectral number fails the promotion test: the latter is $\gamma_{\text{repair step}}$ until the active fiber, physical clock, and common-parent response pole are checked. If the exchange branch is nonzero, the source packet must also carry explicit recipient stress and equal-and-opposite exchange current, and all source/solver/likelihood hashes must be immutable before data comparison. The consensus-protocol paper supplies the relevant test language and fixed-point firewall; this compact proof cites that package instead of reproducing the full math [A23,A24,A25].

Collider tests [O5,C2,S6]. The OPH particle files contain concrete collider-facing numbers, including $m_H = 125.1995304097179\,\mathrm{GeV}$, $m_t = 172.35235532883115\,\mathrm{GeV}$, $M_W = 80.377\,\mathrm{GeV}$, and $M_Z = 91.18797809193725\,\mathrm{GeV}$, with the scheme and evidence caveats in [O5,C2]. OPH also predicts no low-scale visible gauge branch incompatible with the recovered $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y/\mathbb{Z}_6$ quotient [O4,O5]. Falsification: scheme-controlled collider measurements exclude the stated mass surfaces, or CERN/FCC discovers a light visible gauge boson or chiral exotic incompatible with the OPH quotient branch.

χ_ν lift response [O8,O7,C4]. The scalar-channel identity uses the coherent-matter same-channel theorem in [O3,O8]: a nondegenerate OPH-coherent material source

$$S_{\mathrm{coh}}^{\mathrm{can}} = \mathbf{1}_{\text{self-read}} R_U P_U C_U$$

lies in the same edge-center scalar channel under Scalar Edge-Center Exhaustion. The coefficient row listed in [O8] uses the public comparison coordinate P_C: $\chi_\nu^{\mathrm{can}} = e^{-P_C/24} = 0.9343006394893864\ldots$, with $0.9343006394893864 \le \chi_\nu^{\mathrm{can}} \le 1$, and $\chi_\nu^{\mathrm{eng}} = \chi_\nu^{\mathrm{can}}/N_{\mathrm{coh}}$. For $N_{\mathrm{coh}} = 10^{22}$, this is $9.34 \times 10^{-23} \le \chi_\nu^{\mathrm{eng}} \le 10^{-22}$. A hoverboard-class areal load $\Sigma = 200$ to $600\,\mathrm{kg\,m^{-2}}$ needs a vertical canonical coherence contrast $\Delta S_{\mathrm{coh}}^{\mathrm{can}} \simeq 2 \times 10^{-8}$ to 6×10^{-8} for full response. Test: build a substrate that logs ΔS_{coh}, $A_\perp$, Σ, force, temperature, acoustic drive, EM pickup, and mechanical coupling. A null run gives

$$|\chi_\nu^{\mathrm{eng}}| \le \frac{4\pi G F_{\min}}{g^2 A_\perp |\Delta S_{\mathrm{coh}}^{\mathrm{eng}}|}.$$

This row supplies neither finite covariant dark stress nor a macroscopic force receipt. Falsification: "hoverboard doesn't work" when the specified contrast is produced and the control channels stay null.

Dark-sector cosmology [O7,C4]. The diagnostic branch records $\Omega_A = 0.264114401$, $\Omega_m^{\mathrm{CAMB}} = 0.315905206$, $\sigma_8 = 0.807787204$, $S_8 = 0.828924037$, and $\rho_A/\rho_b = 5.363470441$. Test: first emit a finite covariant collar-packet parent with stress closure, recipient stress and exchange-current closure for any nonzero exchange branch, gauge-independent B_A, active-fiber/physical-clock receipts for any promoted Γ_{rec}, refinement convergence, and frozen likelihood hashes; then run the public dark-sector scorecard in [C4] and replace the compact CAMB/SPARC scaffolds by the full covariance likelihoods named in [O7]. Falsification: the fixed branch fails the joint CMB, BAO, SPARC, cluster, and weak-lensing likelihood.

Neutrino closure [O5,C2]. The prediction-escrow section freezes the weighted-cycle masses, splittings, and phases. Test: combine cosmological mass-sum constraints, beta endpoint data, oscillation fits, and neutrinoless-double-beta limits. Falsification: the absolute mass scale or phase-sensitive data exclude the branch. Fastest kill tests are listed in the falsifiability map: $\sum m_\nu = 0.0900119296\,\mathrm{eV}$, beta endpoint $m_\beta \simeq 0.0196\,\mathrm{eV}$, $\Delta m_{21}^2 = 7.4880594651 \times 10^{-5}\mathrm{eV}^2$, $\Delta m_{32}^2 = 2.4374312781 \times 10^{-3}\mathrm{eV}^2$, the listed mixing angles, and $\delta = 305.581^\circ$.

Appendix A. Assigning a number to Newton's constant

The geometric gravity row in the compact argument is

$$G_{\mathrm{geom}} = \ell_\star^2.$$

The SI decimal is a separate clock translation. It counts only when the cesium gap is emitted without measured G, Planck units, or gravity-calibrated proxies [O4,A6]:

$$\varepsilon_{\mathrm{Cs}}^{\mathrm{calc}} = \mathcal{H}_{\mathrm{Cs}}\left(\alpha, \frac{m_e}{E_\star}, \frac{\Lambda_{\mathrm{QCD}}}{E_\star}, \{y_q\}, \mathcal{N}_{133}, \mathcal{B}_{\mathrm{atom}}^{\mathrm{Cs}}\right).$$

If $\varepsilon_{\mathrm{Cs}}$ is back-solved from measured G, the row is a display check. With an independently emitted clock gap, exact SI constants

c, $\hbar$, and $\nu_{\rm Cs} = 9\,192\,631\,770\ \mathrm{s}^{-1}$ give [S3,S4]

$$G_{\rm SI} = \frac{c^5}{4\pi^2\hbar\nu_{\rm Cs}^2}\varepsilon_{\rm Cs}^2 = 6.674299995910528\ldots \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}.$$

The finite-lattice QCD/hadronic part of the cesium calculation is infeasible on conventional von Neumann hardware at the required resolution. OMEGA targets that fixed-point compute class [S14]. This appendix is included for orientation and sits outside the compact anti-circularity proof.

References

Data and external references

[S1] CODATA value for Newton's constant G.
[S2] CODATA inverse fine-structure constant α^{-1}.
[S3] CODATA exact speed of light c.
[S4] CODATA reduced Planck constant $\hbar$.
[S5] Planck Collaboration cosmological parameters.
[S6] Particle Data Group gauge and Higgs-boson summary tables.
[S7] Particle Data Group quark summary tables.
[S8] Particle Data Group lepton summary tables.
[S9] NuFIT 6.0 neutrino oscillation fit.
[S10] KATRIN direct neutrino-mass result.
[S11] Particle Data Group neutrino-mass review.
[S12] PIXEL optical verification plan.
[S13] IBM quantum-cloud OPH experiment code and raw results.
[S14] OMEGA optical fixed-point compute hardware.
[S15] The Right String Theory, OPH blog reference writeup.
[S16] Douglas R. Hofstadter, *I Am a Strange Loop*, Basic Books.

OPH papers

[O1] Observers Are All You Need.
[O2] Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics.
[O3] Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH.
[O4] Recovering Relativity and the Standard Model from Observer Overlap Consistency.
[O5] Deriving the Particle Zoo from Observer Consistency.
[O6] Fine-Structure Constant Derivation.
[O7] Observer-Patch Holography and the Dark Matter Phenomenon.
[O8] Theoretical Bounds on χ_ν in Observer-Patch Holography.
[O9] Observer Patch Holography as String Vacuum Selector.
[O10] Yang-Mills Gap and the Clay Problem.

Code and supporting data

[C1] Pixel-constant derivation code.
[C2] Particle-branch code and supporting files.
[C3] Hierarchy checks and local/global resonance code.
[C4] Dark-sector scorecard and continuation code.
[C5] Consensus and fixed-point architecture code.

Direct reading anchors

Fast lookup: axioms [A1], consensus repair [A2], simulation criterion [A23], consensus refinement/firewall [A24], conditional screen-spectrum gate [A25], pixel closure [A4], capacity closure [A5], hierarchy witness [A10], Standard Model quotient [A21].

[A1] OPH axioms.
[A2] Consensus and repair normal form.
[A3] Patch-carrier and screen microphysics.
[A4] Pixel fixed-point theorem form.
[A5] Capacity fixed point $N_\star$.
[A6] No-G clock requirement.
[A7] Scale relation and Newton-coupling display.
[A8] Unified-coupling interval check $\alpha_U(P)$.
[A9] Electroweak bridge boundary.
[A10] Hierarchy/naturality proof stack.
[A11] 12-port screen sieve.
[A12] Fine-structure trunk.
[A13] Public endpoint caveat.
[A14] Support boundary.
[A15] Koide charged-family roots.

[**A16**] Charged-family shape boundary.
[**A17**] Neutrino weighted-cycle theorem branch.
[**A18**] Born-Lueders central-record theorem.
[**A19**] Bell/CHSH Tsirelson event surface.
[**A20**] Product-group consequences and no X/Y proton-decay carrier.
[**A21**] Standard Model quotient and $\mathbb{Z}_6$ kernel.
[**A22**] Edge-entropy/area law identifies the Einstein/Newton coupling and derives the factor of 4.
[**A23**] OPH simulation test, fixed-point firewall, and falsifiers [O2].
[**A24**] Finite and refinement consensus theorem package, including quotient closure, coarse-graining compatibility, Markov-collar splice, and downstream test boundary [O2].
[**A25**] Conditional OPH screen-spectrum theorem and firewall: screen covariance/tilt are conditional; scalar release amplitude, source provenance, pooled reducers, radial lift, and physical TT/TE/EE spectra remain separate gates [O4].